

Partially Ordered Stochastic Conformance Checking

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1 Scenario

We use the toy Petri net as the running example throughout this report (Figure 1) to illustrate the concepts in Leemans et al. [1]. After $T1$, the process splits into two concurrent branches: $T2$ and $T3$ are independent and may occur in either order. Both branches must complete before $T4$ can fire. From there, execution either continues to $T6$ (termination) or takes $T5$, which loops back and enables another iteration of the same concurrent block.

This simple example already exhibits the two challenges that motivate partially ordered stochastic conformance checking: First, concurrency creates multiple equivalent interleavings that represent the same partial order. And second, the loop can be repeated arbitrarily many times leading to an infinite stochastic language for the model.

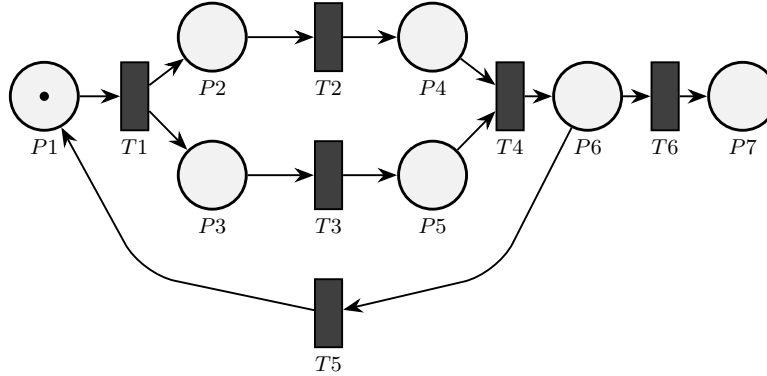


Fig. 1. Example Petri net with concurrency on $\{T2, T3\}$ and a loop enabling repeated executions via $T5$.

2 Technique

This section summarizes the technique proposed by Leemans et al. [1]. Leemans et al. introduce a novel approach to stochastic conformance checking (SCC) that

accounts for concurrency in process models and logs. The technique is based on the concept of partially ordered traces (PO-traces) and a two-layer distance calculation using the Earth Movers’ Stochastic Conformance (EMSC) framework introduced in [3].

2.1 Setting

Leemans et al.’s approach considers conformance checking between an event log (observed behavior) and a process model (ideal behavior), where both are represented as stochastic languages over partially ordered traces. Both model and behavior are interpreted as probability distributions over possible executions. Conformance is defined as the distance between these distributions, measured using EMSC. The model is assumed to be a Labeled Stochastic Petri Net (LSPN), like the one shown in Figure 1.

The technique is only applicable to models with the following properties:

- Safety: A model is safe if no place can contain more than one token ($\forall p \in P : M(p) \leq 1$). This simplifies the computation of run probabilities.
- Sound: Models are sound if they can always reach a proper completion state, that is, a state in which only the designated end place (P7 in our scenario) holds a token and no transitions are enabled. This guarantees that runs don’t get stuck in deadlocks, which would cause leakage of probability mass, making the stochastic language ill-defined.
- Confusion-freeness: If two transitions share an input place but do not share all input places, they cannot be jointly enabled in any reachable marking. This guarantees that concurrent transitions are independent and allows us to disregard their execution order when computing probabilities.

The output of the technique is a distance value that quantifies how well model and log conform.

2.2 Partial-Order Traces and Their Linearizations

Each run of an LSPN corresponds to a total-order trace (TO-trace) that lists the sequence of transition firings. However, when transitions are concurrent, multiple TO-traces can represent the same underlying behavior. To capture this, the technique uses partially ordered traces (PO-traces), which allow for sets of concurrent events to be represented without committing to a specific order. For example, in our scenario, transitions $T2$ and $T3$ are concurrent, so the PO-trace $\langle T1, \{T2, T3\}, T4, T6 \rangle$ represents both total-order traces $\langle T1, T2, T3, T4, T6 \rangle$ and $\langle T1, T3, T2, T4, T6 \rangle$. More generally, the set of total-order traces consistent with a PO-trace ρ is its set of linearizations, denoted $\text{lin}(\rho)$. To model stochastic behavior, probabilities are assigned directly to PO-traces. Formally, a stochastic PO-language is a function $L : \mathcal{P} \rightarrow [0, 1]$ with $\sum_{\rho \in \mathcal{P}} L(\rho) = 1$, where $\mathcal{P}$ is the set of PO-traces. This gives a probability distribution over partially ordered behavior, which is the input representation used in the conformance metric of the technique.

2.3 Two-Layer Distance Calculation

Leemans et al.'s conformance score over the stochastic languages of the model and log is computed in two layers. First, the method compares individual executions at the trace level using Levenshtein distance $\delta(\sigma, \sigma')$. Second, it then compares the full stochastic behavior at the language level by aggregating these trace-level differences with probability weights using Earth Movers' Stochastic Conformance (EMSC) [3]. Colloquially, the Levenshtein distance expresses how different two traces are and the EMSC distance expresses how probability mass is distributed across these differences.

Per-Trace Distance: Levenshtein Distance At the trace level, the method compares two total-order traces σ and σ' using the Levenshtein distance $\delta(\sigma, \sigma')$: the minimum number of insertions, deletions, and substitutions needed to transform one trace into the other. This distance can be interpreted in terms of alignment moves: a synchronous move (matching events $e = e'$) has a cost of 0, while log-only or model-only moves (insertions/deletions, $e \neq e'$) and substitutions each have a cost of 1.

EMSC Distance over Stochastic Languages At the language level, Earth Movers' Stochastic Conformance (EMSC) compares two stochastic languages L and L' , by determining how much probability mass must be moved between them to transform one into the other. This probability mass is weighted by the trace level distance. Here, ρ and ρ' denote PO-traces (elements of $\mathcal{P}$), while σ and σ' denote total-order traces (linearizations).

The paper differentiates two variants of EMSC with respect to assumptions about the log's concurrency. In both cases, the model is assumed to be certain:

- **Certain concurrency (EMSC-CC)** assumes that all linearizations of each of the models PO-traces are truly concurrent. The distance therefore is defined as the best case (minimum) trace distance over all linearization pairs.

$$\Delta_{CC}(\rho, \rho') = \min_{\substack{\sigma \in \text{lin}(\rho) \\ \sigma' \in \text{lin}(\rho')}} \delta(\sigma, \sigma')$$

Given this PO-trace distance, the language-level EMSC-CC distance/conformance score is computed as:

$$\text{EMSC}_{CC}(L, L', f) = \sum_{\substack{\rho \in L \\ \rho' \in L'}} f(\rho, \rho') \cdot \Delta_{CC}(\rho, \rho')$$

, where $f(\rho, \rho')$ denotes the amount of probability mass moved from ρ to ρ' .

- **Uncertain concurrency (EMSC-UC)** assumes the log's concurrency stems from uncertain observations as opposed to being truly concurrent. Many linearizations of its PO-traces are possible, but only one actually occurred.

Therefore, conformance is defined as a range between the best case (minimum) and worst case (maximum) trace distance over all linearization pairs:

$$\begin{aligned}\Delta_{UC\text{worst}}(\rho, \rho') &= \max_{\sigma \in \text{lin}(\rho)} \min_{\sigma' \in \text{lin}(\rho')} \delta(\sigma, \sigma') \\ \Delta_{UC\text{best}}(\rho, \rho') &= \Delta_{CC}(\rho, \rho')\end{aligned}$$

Let $f(\rho, \rho')$ again denote the transported probability mass from ρ to ρ' :

$$\begin{aligned}\text{EMSC}_{UC\text{worst}}(L, L', f) &= \sum_{\substack{\rho \in L \\ \rho' \in L'}} f(\rho, \rho') \cdot \Delta_{UC\text{worst}}(\rho, \rho') \\ \text{EMSC}_{UC\text{best}}(L, L', f) &= \sum_{\substack{\rho \in L \\ \rho' \in L'}} f(\rho, \rho') \cdot \Delta_{UC\text{best}}(\rho, \rho')\end{aligned}$$

Infinite Behavior, Truncation, and Bounds Models with loops, such as our example in Figure 1, have infinite stochastic languages, so it is not feasible to compute the exact EMSC over all PO-traces. The method therefore truncates the model language to its top- q most probable PO-traces as well as r random PO-traces. This ensures our distance metrics are well defined and computable. Let $\tilde{L}'$ denote this truncated model language. The omitted probability mass is tracked as

$$L'(\perp) = 1 - \sum_{\rho \in \tilde{L}'} L'(\rho).$$

This truncation introduces bounds to EMSC_{CC} and widens the range of the existing bounds of EMSC_{UC} . With higher q and r , more of the model's behavior is captured, leading to narrower bounds. In the limit $q, r \rightarrow \infty$, the bounds converge to the conformance value of the full (untruncated) language.

3 Advantages & Disadvantages

3.1 Advantages

Compact representation. One PO-trace can represent many total-order traces. For example, in our scenario, the PO-trace $\langle T1, \{T2, T3\}, T4, T6 \rangle$ represents 2 TO-traces. With each loop iteration, the number of TO-traces corresponding to a single PO-trace doubles, so with i loop iterations this compresses 2^i TO-traces into a single PO-trace:

$$\rho_{\text{PO}} = \underbrace{\langle T1, \{T2, T3\}, T4 \rangle}_{2 \text{ total-order traces}}, \underbrace{\langle T5, T1, \{T2, T3\}, T4 \rangle^*}_{2^{i-1} \text{ total-order traces}}, T6$$

This reduces the number of traces to compare at the language level, which is crucial for scalability.

Concurrency semantics. PO-traces treat concurrent transitions as unordered, so conformance is not affected by arbitrary interleavings (noise). This is faithful to the behavior of concurrent processes, where the execution order of concurrent events is irrelevant. In contrast, TO-traces treat different interleavings as different behavior. In practice, this makes TO-trace-based approaches sensitive to timestamping noise and other sources of spurious ordering, while PO-trace-based approaches are less affected by such issues.

Loop bounding. Truncation (keeping top- q runs) makes loop-heavy models computationally tractable while providing formal bounds on the introduced error. As q and r increase, bounds narrow. This enables a qualified trade-off between precision and computational cost, enabling adjustments based on requirements for quality and resources.

3.2 Disadvantages

Computational Bottleneck. The approach reduces interleaving blow-up at the language level, but distance computation between PO-traces remains costly. For each PO-trace pair, the technique may have to reason over many possible linearizations, whose number grows with the number of concurrent events. The paper identifies distance computation, rather than the final transport optimization, as the main bottleneck.

Restrictive Assumptions. The method assumes labeled stochastic Petri nets that are confusion-free, safe, and sound. These assumptions make the run generation and probability calculation tractable, but they restrict the class of admissible input models. The implementation also assumes these properties instead of automatically verifying them, so violations may lead to unreliable results rather than explicit errors.

Inexactness due to Truncation. For uncertain log orderings (EMSC-UC) and/or truncated model behavior, the method returns lower and upper bounds rather than a single exact value. These bounds are useful because they make approximation error explicit, but they may become wide when much probability mass is truncated or when uncertain linearizations differ strongly. In such cases the result can indicate limited precision even though the computation itself succeeds.

4 Positioning

This section positions the PO-trace approach of [1] against two related methods, Enjoy the Silence [2] and Abstract-and-Compare [4]. The core distinction is what each method preserves most strongly. Enjoy the Silence emphasizes exact probability analysis with silent transitions, PO-traces emphasize explicit concurrency semantics, and Abstract-and-Compare emphasizes robustness and scalability through abstraction.

4.1 Enjoy the Silence: Analysis of Stochastic Petri Nets with Silent Transitions

Paper: Leemans, Maggi, Montali [2].

A main contribution of Enjoy the Silence is exact finite-trace analysis for stochastic Petri nets with silent transitions. It introduces labeled stochastic processes and derives key probabilities through absorbing Markov chain analysis with automata-based filtering. This is important because silent transitions can change visible trace probabilities and therefore affect conformance scores. Relative to PO-traces, the method is stronger on probabilistic foundations with hidden behavior but it does not model the certain-versus-uncertain concurrency distinction used by EMSC-CC and EMSC-UC. It is therefore the better choice when silent-transition fidelity is the main requirement, while PO-traces are better when the primary requirement is explicit treatment of concurrent behavior.

4.2 Abstract-and-Compare Stochastic Conformance Checking

Paper: Goulart Rocha, Leemans, van der Aalst [4].

Abstract-and-Compare addresses a different bottleneck, namely scalability and robustness in large or noisy comparisons. Instead of comparing full stochastic languages directly, it maps model and log to an m^k abstraction based on expected subtrace frequencies and computes conformance on that abstraction. The parameter k sets how much local context is retained. Larger k preserves more structure but increases computational cost. This strategy smooths partial mismatches and can improve robustness, but it does not preserve explicit partial-order semantics. In the running example, uncertain ordering between $T2$ and $T3$ is represented explicitly in the PO-trace approach by best-case and worst-case bounds, while Abstract-and-Compare captures the effect only indirectly through subtrace statistics.

Overall, the three approaches define a clear choice space for stochastic conformance checking. Enjoy the Silence is preferable when exact probability treatment of silent transitions is essential. PO-traces are preferable when concurrency semantics and EMSC-CC or EMSC-UC interpretation must remain explicit. Abstract-and-Compare is preferable when scalability and robustness to partial mismatch are the dominant practical goals.

References

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